

SDAM first-level analysis with DKGE

Design-Kernel Group Embedding on MVPA AUC maps

dkge example

2026-05-09

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1 Dataset and study design

The dataset is 32 participants (14 controls, 18 SDAM) each contributing four whole-brain MVPA AUC maps in a 2 x 2 within-subject design crossing **task** (**recog**, **nback**) with **measure** (**low_mid**, **high_sem**). Maps are AUC - 0.5 so they are zero-centred at chance. The shared analysis mask contains 218,414 voxels where every map is finite and nonzero.

Table 1: Four cells of the within-subject 2 x 2 design.

task	measure	cell label
recog	low_mid	recog_low_mid
recog	high_sem	recog_high_sem
nback	low_mid	nback_low_mid
nback	high_sem	nback_high_sem

2 DKGE setup

Each subject contributes a $q \times P_s$ matrix B_s whose rows are the four cell maps in canonical order. Because each cell is observed once per subject, the per-subject “design” is the $q \times q$ identity. The factorial structure is encoded by the design kernel:

```
kern <- make_sdam_design_kernel(rho_task = 1, rho_measure = 1, rho_int = 1)
round(kern$K, 3)
```

	recog_low_mid	recog_high_sem	nback_low_mid	nback_high_sem
recog_low_mid	0.250	0.083	0.083	0.000
recog_high_sem	0.083	0.250	0.000	0.083
nback_low_mid	0.083	0.000	0.250	0.083
nback_high_sem	0.000	0.083	0.083	0.250

Diagonal cells are similar to themselves; off-diagonal entries are nonzero when two cells share **task** or **measure**, encoding factorial similarity that classical PLS cannot represent.

3 Component summary

Table 2: Variance carried by each DKGE component ($q = 4$, kept rank = 3).

component	sdev	variance	variance_percent	cumulative
1	69.2556	4796.3439	78.89	78.89
2	28.8187	830.5203	13.66	92.55
3	21.2835	452.9890	7.45	100.00

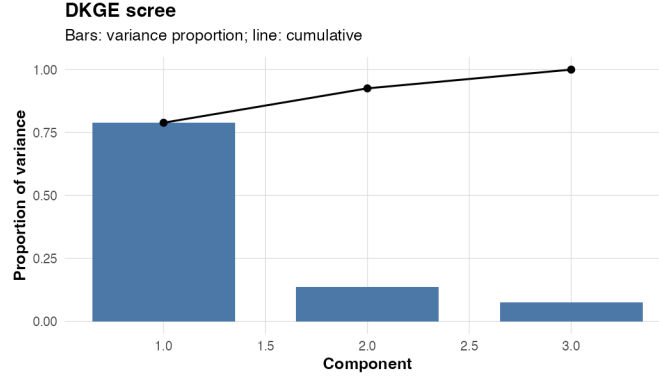


Figure 1: Scree plot of the DKGE compressed-covariance eigenvalues.

4 Design saliences

The cell-space loading matrix $K \% \% U$ is the DKGE analogue of PLS design saliences. Each panel below is one component; cells are laid out as **task** x **measure** so that main effects and interaction structure are visible at a glance.

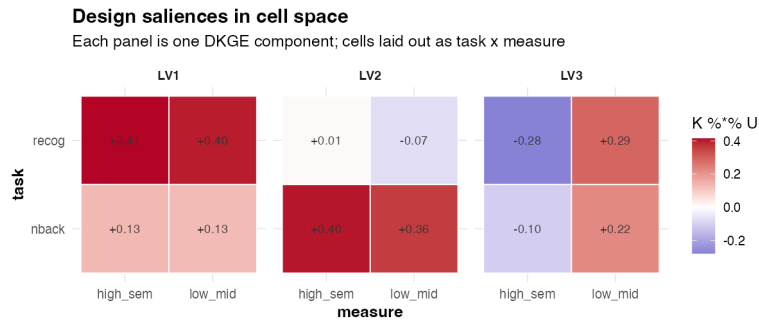


Figure 2: Cell-space loadings for each component.

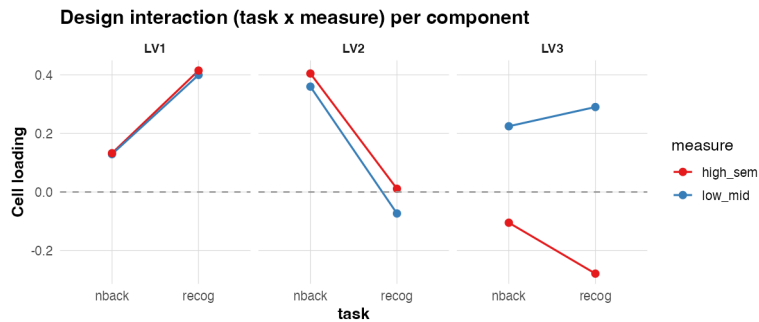


Figure 3: Interaction view: lines connect cells sharing a measure across tasks. Parallel lines indicate additive structure; crossing lines indicate task x measure interaction.

5 Subject-level diagnostics

DKGE's MFA-style subject weighting downweights noisy subjects automatically. Energy on each component shows where each subject's variance lands.

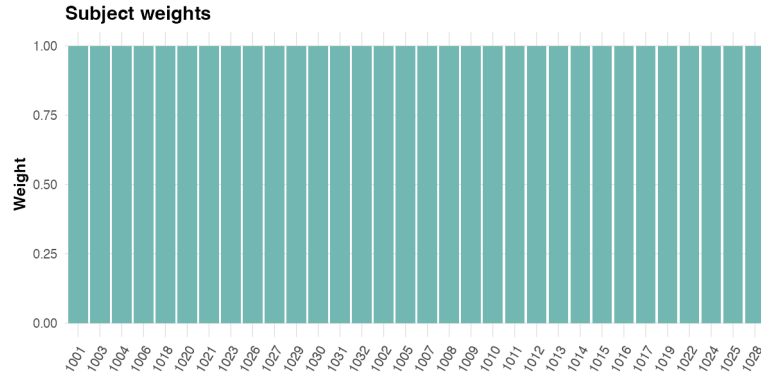


Figure 4: Subject weights and per-component participation energy.

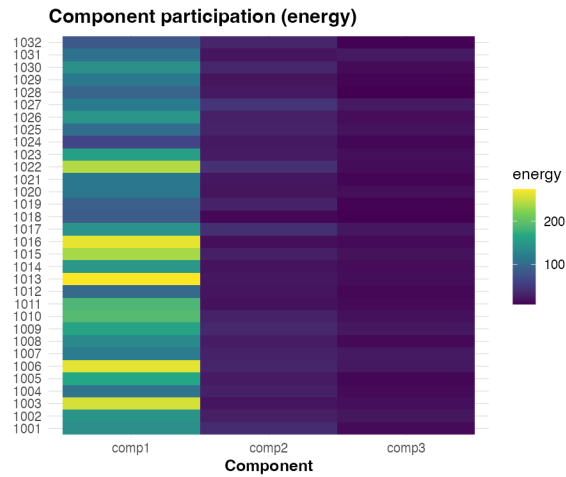


Figure 5: Subject weights and per-component participation energy.

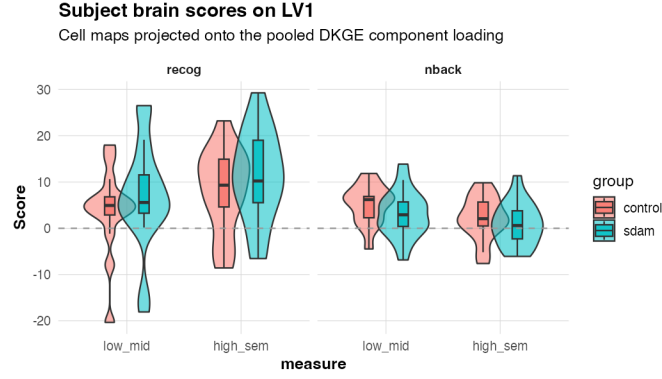


Figure 6: Subject ‘brain scores’ on LV1, by task (facets) x measure (x axis) and group (fill). Equivalent role to PLS scores plotted by condition.

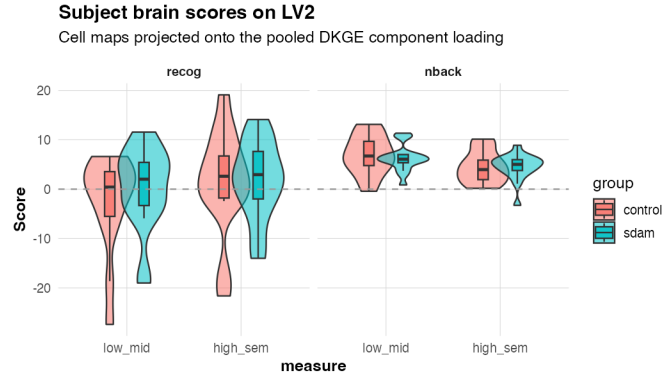


Figure 7: Subject brain scores on LV2.

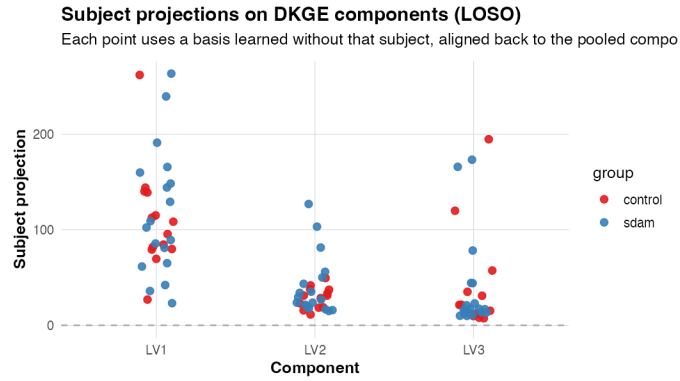


Figure 8: LOSO subject projections on each DKGE component. Each point is one subject; the basis used for that point was learned without that subject, then aligned back to the pooled component axes.

The score plots project each subject-cell map onto a pooled DKGE component loading map, so they are the DKGE score-space analogue of PLS brain-score plots. The table below tests whether group differs on each component-level cell contrast.

Table 3: Group differences in DKGE component score-space contrasts. The task, measure, and task:measure rows are the group-by-within-subject effects for each component.

	component	contrast	mean_control	mean_sdam	diff_sdam_minus_control	statistic	p	p_adjusted
7	LV1	task	-2.3776	-7.1150	-4.7373	1.4497	0.1575	0.8922
4	LV1	measure	0.7712	1.0590	0.2877	-0.1904	0.8508	0.8961
10	LV1	task:measure	-3.6977	-3.1844	0.5133	-0.3305	0.7435	0.8922
1	LV1	grand_mean	4.8091	5.5157	0.7067	-0.3655	0.7173	0.8922
8	LV2	task	6.3149	4.4220	-1.8929	0.6283	0.5357	0.8922
5	LV2	measure	0.7039	-0.1168	-0.8207	0.6567	0.5166	0.8922
11	LV2	task:measure	-3.1438	-1.6240	1.5199	-1.2111	0.2376	0.8922
2	LV2	grand_mean	2.5577	3.1896	0.6319	-0.3490	0.7301	0.8922
9	LV3	task	0.6559	1.3613	0.7054	-1.3726	0.1807	0.8922
6	LV3	measure	-0.9483	-1.0511	-0.1028	0.3794	0.7077	0.8922
12	LV3	task:measure	0.6104	0.7098	0.0994	-0.4406	0.6627	0.8922
3	LV3	grand_mean	-0.1255	-0.0908	0.0347	-0.1317	0.8961	0.8961

6 LOSO contrast maps

Cross-fitted (leave-one-subject-out) contrast values give an unbiased per-subject voxel map for each design contrast. The summary statistics below are computed across the 32 subjects.

Table 4: Voxel statistics for each LOSO contrast: mean across subjects, $z = \text{mean} / \text{SE}$, and the number of voxels with $|z| \geq 3$.

contrast	mean_min	mean_max	z_min	z_max	n_z_ge3
task	-0.007	0.011	-4.83	5.98	5136
measure	-0.002	0.004	-4.78	4.82	1906
task:measure	-0.001	0.001	-4.85	4.77	2263
grand_mean	-0.004	0.012	-3.54	6.12	5548

7 Bootstrap-ratio brain maps

For each contrast we resample the 32 LOSO subject vectors with replacement (200 draws) and report $z = \text{mean} / \text{SD}$ at every voxel. This is the DKGE analogue of the PLS bootstrap-ratio map.

Table 5: Bootstrap z -statistic per voxel for each contrast.

contrast	min_z	max_z	pos_ge3	neg_le_neg3
task	-5.42	6.17	4007	1646
measure	-5.12	5.25	2575	164
task:measure	-5.28	5.08	158	3065

Task contrast (nback - recog) BSR z-map

$|z| \geq 3$ shown; q-space bootstrap of contrast

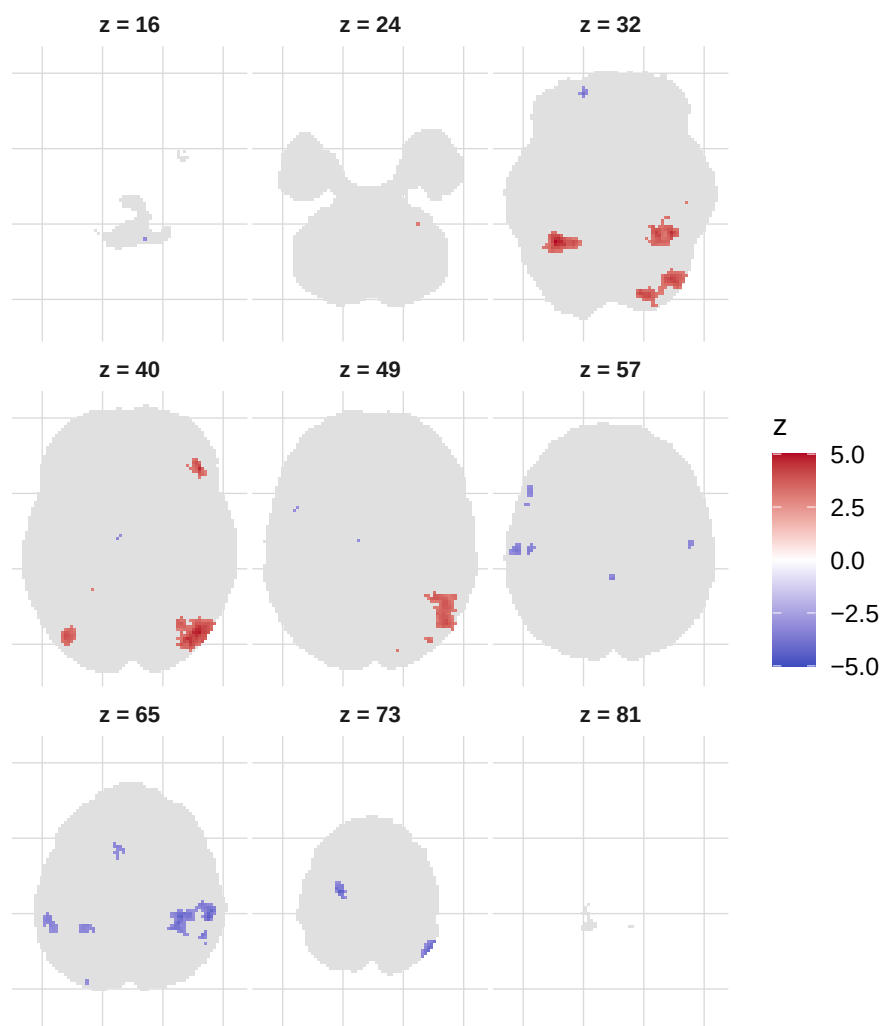


Figure 9: Task contrast (nback - recog) bootstrap z-map; nine evenly-spaced axial slices, $|z| \geq 3$.

Measure contrast (high_sem - low_mid) BSR z-map

$|z| \geq 3$ shown; q-space bootstrap of contrast

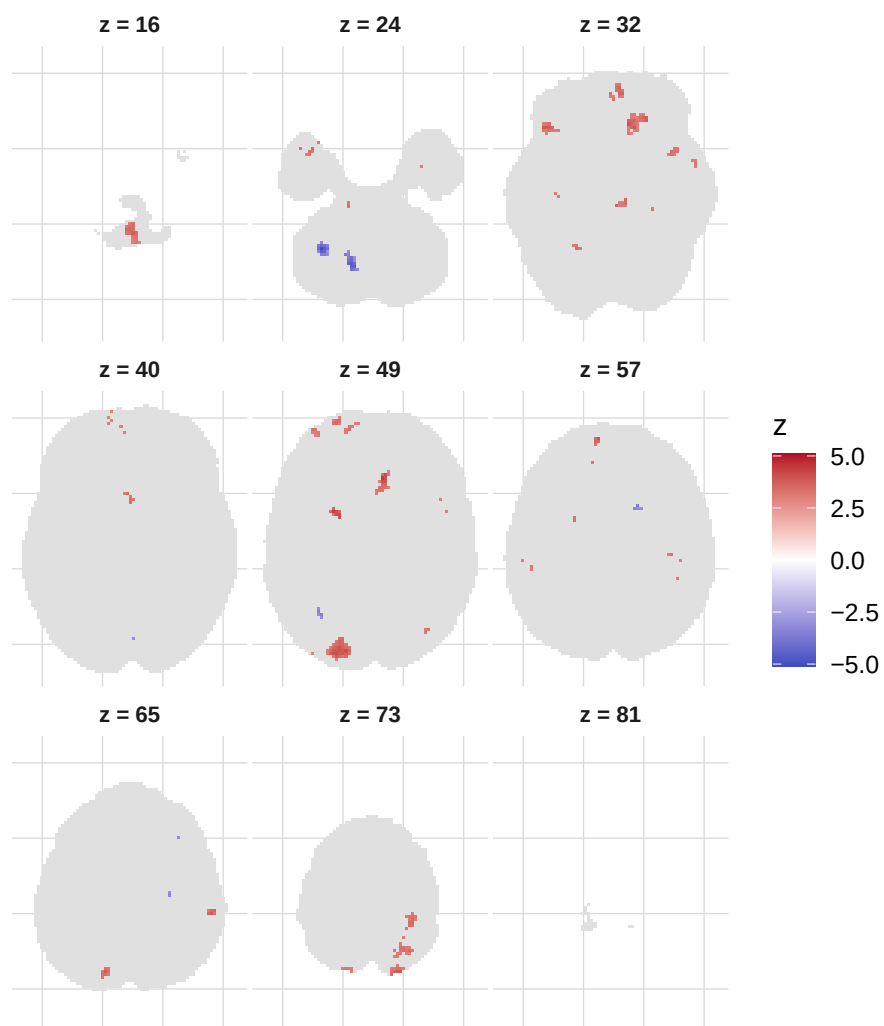


Figure 10: Measure contrast (high_sem - low_mid) bootstrap z-map; nine evenly-spaced axial slices, $|z| \geq 3$.

Task x measure interaction BSR z-map

$|z| \geq 3$ shown; q-space bootstrap of contrast

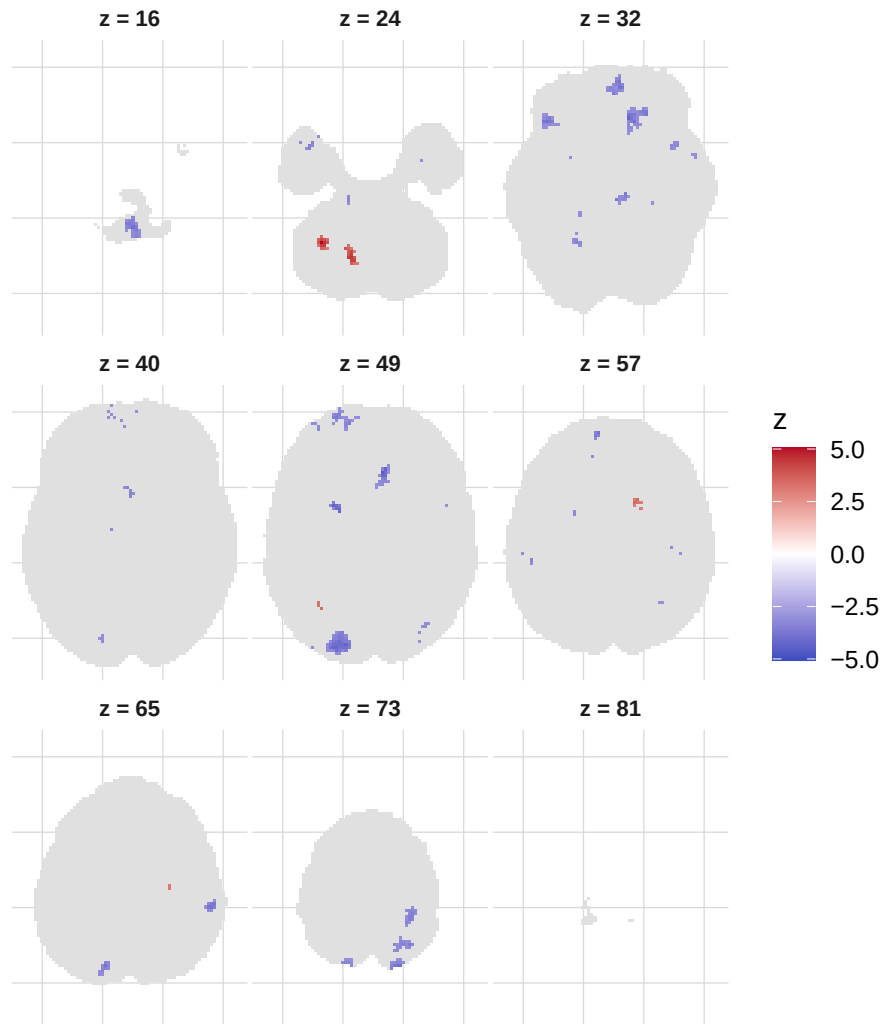


Figure 11: Task x measure interaction bootstrap z-map; nine evenly-spaced axial slices, $|z| \geq 3$.

DKGE bootstrap z-map

$|z| \geq 3$; q-space bootstrap, voxel projection

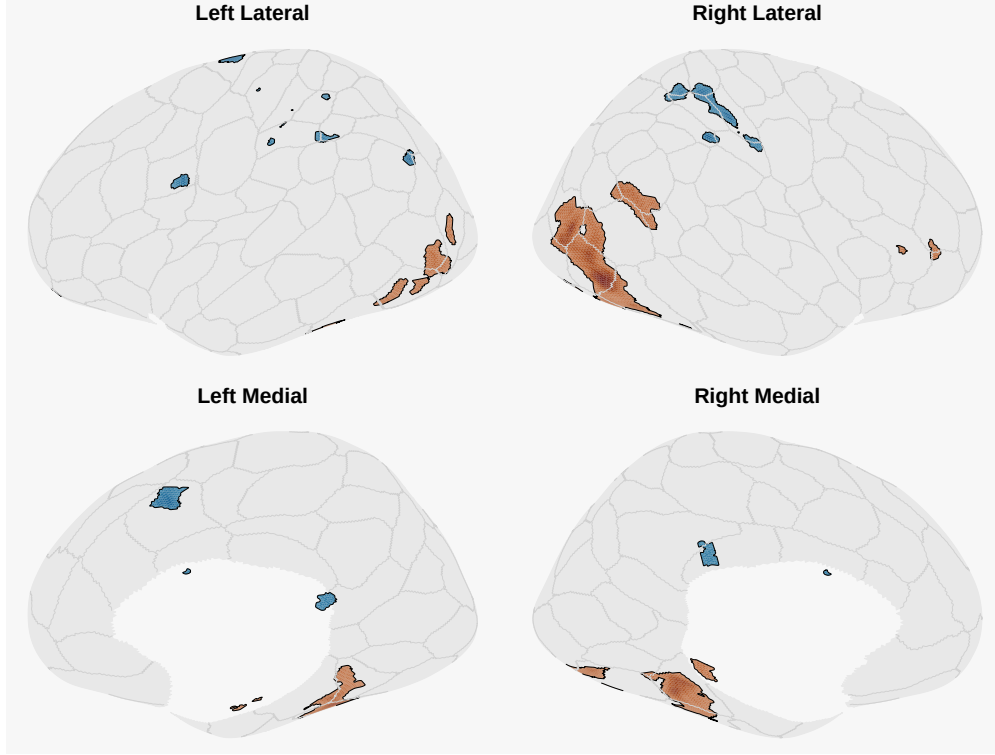


Figure 12: Task contrast bootstrap z-map projected onto a Schaefer 200x7 surface (fsaverage6).

8 Primary between-subject DKGE: control vs SDAM

The voxel-level contrast-map target Y is regressed on $\sim \text{group}$ via reduced-rank regression. Rows are subjects; columns are feature blocks formed by each cross-fitted repeated-measures contrast (`grand_mean`, `task`, `measure`, `task:measure`) crossed with voxels. This tests whether control and SDAM subjects differ in the spatial expression of the within-subject effects, rather than reducing each subject to three component means first.

This is the primary DKGE inference path for the PLS-like crossover question: fit a shared $q = 4$ task x measure DKGE model, build cross-fitted subject-level contrast targets, and test `group` as a between-subject predictor.

Table 6: Between-subject target dimensions: 32 subjects x 873,656 contrast-voxel features.

contrast	features
task	218414
measure	218414
task:measure	218414
grand_mean	218414

Table 7: Permutation test for the group term in the voxel-level contrast-map RRR.

	term	statistic	p	p_adjusted
group	group	0.4882	0.495	0.495

Table 8: Focused primary test: group effect on the cross-fitted task:measure contrast target only.

	term	statistic	p	p_adjusted	target_features
group	group	3e-04	0.89	0.89	218414

Table 9: Top featurewise group effects by maxT-adjusted p-value.

	contrast	feature	statistic	p	p_adjusted
723733	grand_mean	grand_mean:68491	0.0028	0.528	0.528
723774	grand_mean	grand_mean:68532	0.0028	0.526	0.531
723694	grand_mean	grand_mean:68452	0.0027	0.520	0.534
723732	grand_mean	grand_mean:68490	0.0027	0.521	0.539
723693	grand_mean	grand_mean:68451	0.0027	0.520	0.543
718764	grand_mean	grand_mean:63522	0.0027	0.521	0.546
728732	grand_mean	grand_mean:73490	0.0027	0.526	0.546
723773	grand_mean	grand_mean:68531	0.0026	0.520	0.548
723731	grand_mean	grand_mean:68489	0.0026	0.520	0.548
723696	grand_mean	grand_mean:68454	0.0026	0.529	0.551

9 $q = 8$ group-cell mean DKGE bridge to PLS

This is the $q = 8$ cell-mean analysis in which **group** enters the analysis up front. It is not the strict partial-effect subject-level fit. The target is an 8-row matrix of observed group-cell means: for each group, task, and measure row, subject maps are averaged within group before the decomposition.

Mean-centered task PLS decomposes this kind of group/condition cell-mean object, so this is the DKGE bridge that is closest to the PLS LV display. We build the 8-row **group x task x measure** target through `dkge_aggregate_target()`, apply a DKGE design kernel over those aggregate rows, and decompose the resulting cell-mean object. It is a secondary PLS-facing analysis; the primary DKGE inference remains the $q = 4$ subject-level model above.

Permutation and bootstrap inference still resample subjects, not the eight aggregate rows. For every draw, the subject labels or subject sample are changed, the group x task x measure means are recomputed, the aggregate decomposition is refit, components are aligned back to the observed fit, and the group:task:measure statistic is reevaluated. The bridge statistic below is a selected-component, singular-value-scaled salience contrast on the cell-mean object. It is useful for comparing the shape of DKGE and PLS cell-mean decompositions, but it is not a replacement for the primary subject-level `dkge_between_*` test above.

Table 10: $q = 8$ group-cell mean DKGE bridge target and fit settings.

rows	features	aggregate	inference_unit	center	rank	tested_component
8	218414	mean	subject	column	3	LV2

Table 11: The eight observed group x task x measure cell means decomposed by the bridge analysis.

group	task	measure
control	nback	high_sem
control	nback	low_mid
control	recog	high_sem
control	recog	low_mid
sdam	nback	high_sem
sdam	nback	low_mid
sdam	recog	high_sem
sdam	recog	low_mid

Table 12: Singular values of the $q = 8$ group-cell mean DKGE bridge.

component	singular_value
LV1	2.1533
LV2	1.4701
LV3	0.9191

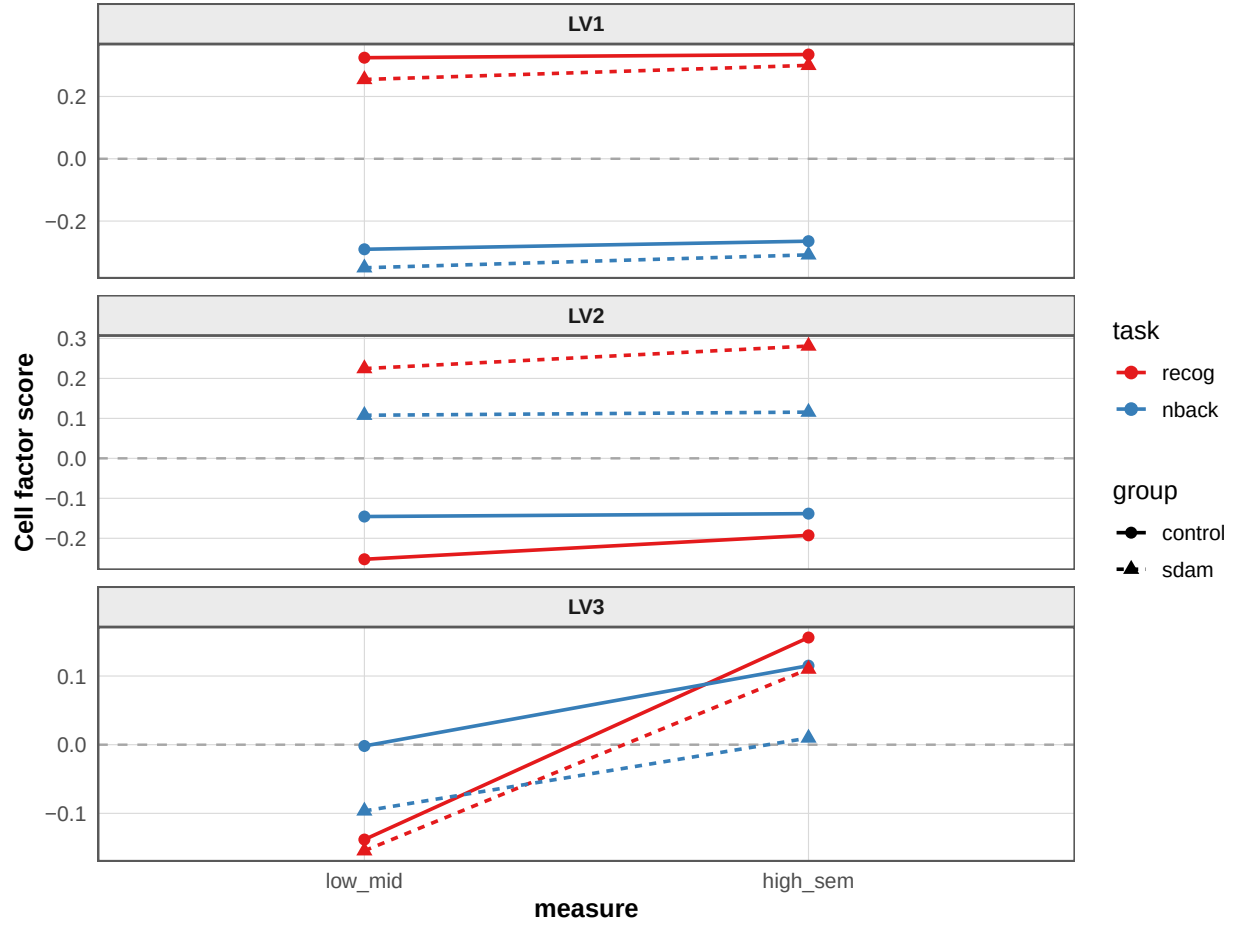


Figure 13: $q = 8$ group-cell mean DKGE bridge factor scores. Each LV has its own boxed panel; task is color and group is line type/point shape.

Table 13: LV-wise $q = 8$ bridge cell-score contrasts from subject bootstrap. Intervals excluding zero mark visually large effects that were stable under subject resampling.

contrast	component	observed	bootstrap_low	bootstrap_high	excludes_zero
task	LV1	-0.6068	-0.8827	-0.0453	TRUE
task	LV2	-0.0302	-0.5867	0.5686	FALSE
task	LV3	0.0132	-0.0854	0.1140	FALSE
measure	LV1	0.0307	-0.2440	0.1837	FALSE
measure	LV2	0.0328	-0.1524	0.1659	FALSE
measure	LV3	0.1955	0.1928	0.3286	TRUE
task:measure	LV1	0.0030	-0.0541	0.1040	FALSE
task:measure	LV2	-0.0253	-0.0663	0.0638	FALSE
task:measure	LV3	-0.0839	-0.1286	-0.0467	TRUE
group	LV1	-0.0520	-0.6887	0.6218	FALSE
group	LV2	0.3645	0.0481	0.6452	TRUE
group	LV3	-0.0656	-0.1680	0.0924	FALSE
group:task	LV1	0.0003	-0.1778	0.2561	FALSE

contrast	component	observed	bootstrap_low	bootstrap_high	excludes_zero
group:task	LV2	-0.1108	-0.1967	0.0032	FALSE
group:task	LV3	-0.0344	-0.1083	0.0250	FALSE
group:measure	LV1	0.0126	-0.0164	0.0472	FALSE
group:measure	LV2	-0.0007	-0.0234	0.0262	FALSE
group:measure	LV3	-0.0101	-0.0623	0.0257	FALSE
group:task:measure	LV1	-0.0049	-0.0140	0.0035	FALSE
group:task:measure	LV2	0.0010	-0.0080	0.0067	FALSE
group:task:measure	LV3	0.0047	-0.0060	0.0173	FALSE

Table 14: Subject-resampling inference for the $q = 8$ group-cell mean DKGE bridge group:task:measure statistic. The statistic is the selected component’s singular-value-scaled salience contrast.

tested_contrast	statistic	permutation	permutation	bootstrap_low	bootstrap_high	bootstrap_peak	B_tie_weak	alignment_diagnostics	alignment_cosine
LV2	0.001	0.718	999	2e-04	0.01	200	0	11	0.2443

Alignment diagnostic: 11 permutation draws had component alignment cosine below 0.50 (minimum 0.244). LV-specific bridge inference should therefore be treated as descriptive unless the statistic is changed to an omnibus/subspace target.

Cell-mean bridge LV1 feature bootstrap z-map

$|z| \geq 2$ shown; subject-resampled cell-mean bridge feature bootstrap

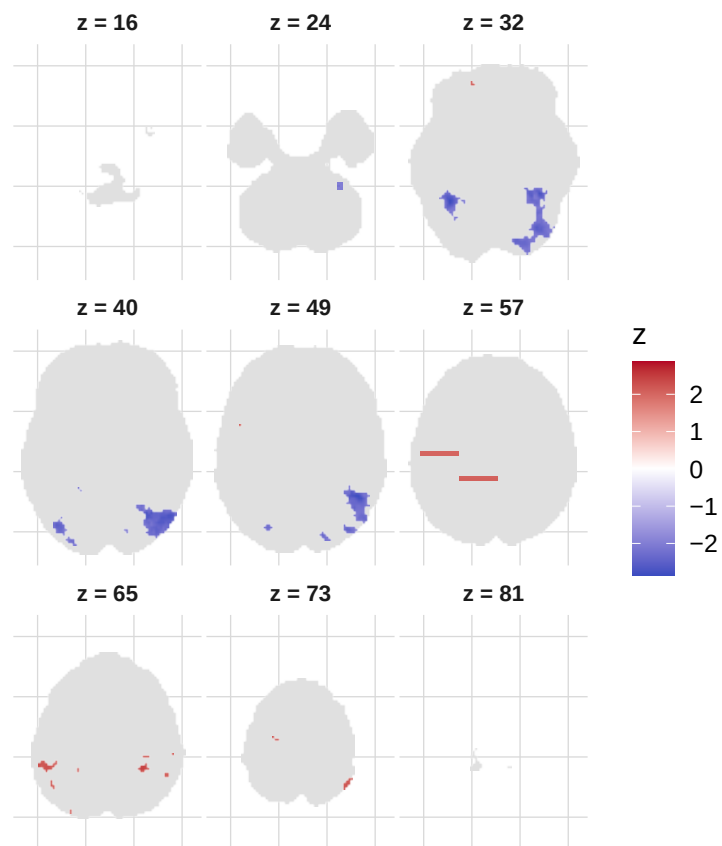


Figure 14: Bootstrap-ratio feature maps for the $q = 8$ group-cell mean DKGE bridge LVs. Maps are built by resampling subjects, recomputing group-cell means, refitting, aligning components, and accumulating feature-space component scores.

Cell-mean bridge LV2 feature bootstrap z-map

$|z| \geq 2$ shown; subject-resampled cell-mean bridge feature bootstrap

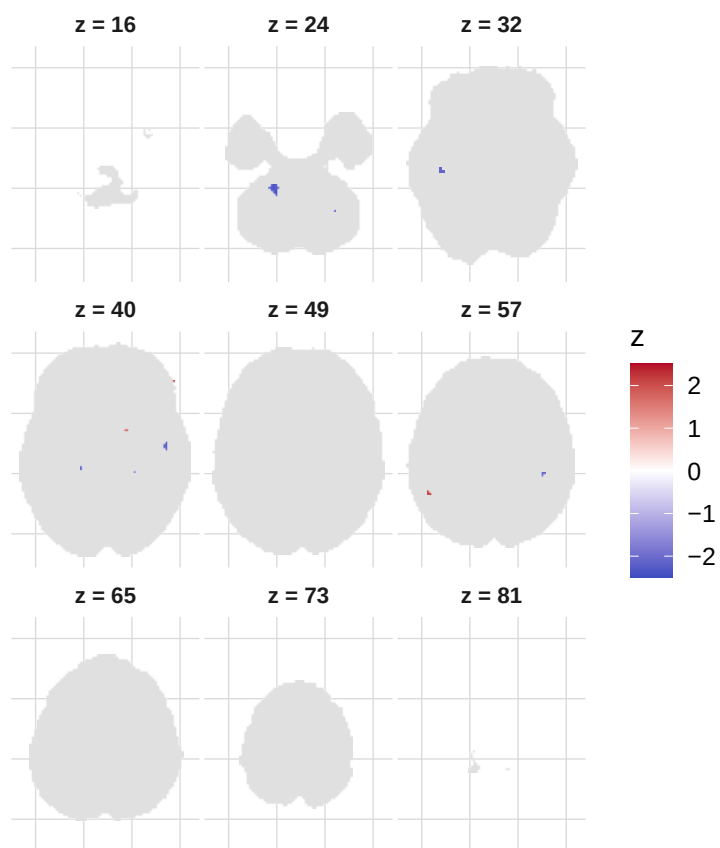


Figure 15: Bootstrap-ratio feature maps for the $q = 8$ group-cell mean DKGE bridge LVs. Maps are built by resampling subjects, recomputing group-cell means, refitting, aligning components, and accumulating feature-space component scores.

Cell-mean bridge LV3 feature bootstrap z-map

$|z| \geq 2$ shown; subject-resampled cell-mean bridge feature bootstrap

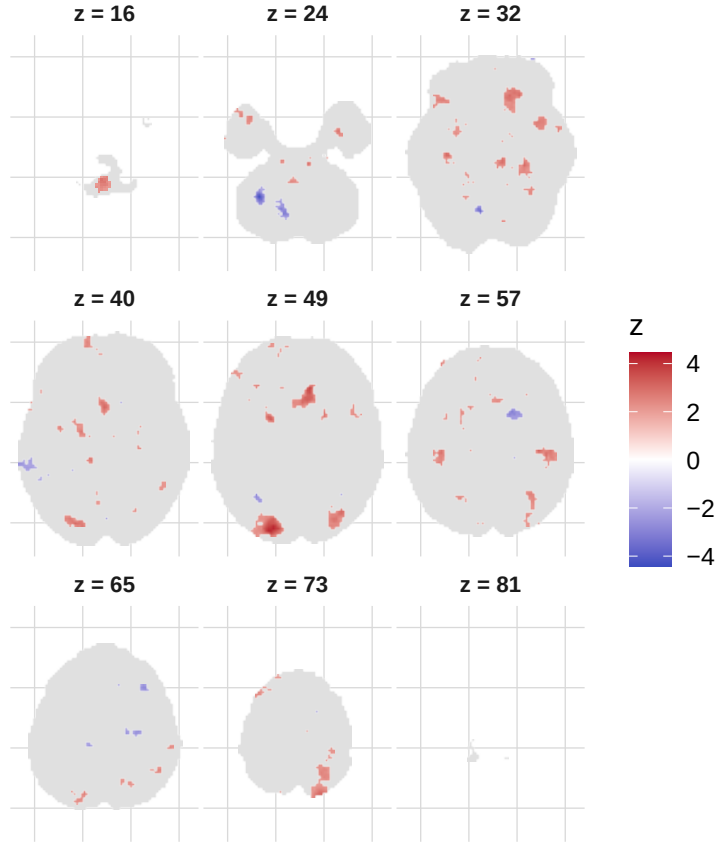


Figure 16: Bootstrap-ratio feature maps for the $q = 8$ group-cell mean DKGE bridge LVs. Maps are built by resampling subjects, recomputing group-cell means, refitting, aligning components, and accumulating feature-space component scores.

10 Strict block-diagonal $q = 8$ partial-effect fit

The strict block-diagonal $q = 8$ partial-effect fit is intentionally omitted from the main figures. In that model each subject observes only the four rows for their own group, and the block-diagonal group kernel leaves the control and SDAM row spaces disconnected. Its components are therefore group-local axes sorted by eigenvalue, not shared group $\times$ task $\times$ measure latent variables. Those component plots are not interpretable as a PLS LV2 analogue, so they are not shown here.

11 Stratified basis diagnostic

The voxel-level group test above assumes the pooled cell-space basis is a reasonable shared design space. As a diagnostic, DKGE is also fit separately in controls and SDAM, then compared with

K-metric principal angles. These angles are descriptive; a stronger claim would compare them against within-group split-half or bootstrap stability.

A geometric note on rank choice. Two rank- r subspaces of a q -dimensional ambient space share at least $\max(0, 2r - q)$ dimensions by the dimension formula, so the first that-many principal angles are forced to zero regardless of the data. With $q = 4$ and pooled rank 3, two of the three angles would be geometric necessities. The diagnostic therefore caps the *stratified* rank at $\text{floor}(q / 2) = 2$ so every reported angle is informative; the pooled fit still uses the full requested rank.

Table 15: Principal angles between control-only and SDAM-only DKGE bases ($q = 4$, stratified rank = 2, forced overlap = 0).

component	group_a	group_b	angle_rad	angle_deg
LV1	control	sdam	0.0039	0.23
LV2	control	sdam	0.2561	14.68

12 Voxel-level group difference (sdam - control) on LOSO contrasts

Table 16: Per-voxel two-sample z statistic for the SDAM minus control difference of each LOSO contrast.

contrast	n_sdam	n_control	mean_diff_min	mean_diff_max	z_min	z_max	n_z_ge3
task	18	14	-0.012	0.010	-4.92	4.22	576
measure	18	14	-0.006	0.005	-4.66	5.52	1009
task:measure	18	14	-0.001	0.002	-5.33	4.73	965
grand_mean	18	14	-0.008	0.009	-5.10	3.97	286

13 Reproducibility

R version: R version 4.5.1 (2025-06-13)

DKGE seed: 20260509

Mask voxels: 218414

Subjects (control / sdam): 14 / 18